

Jinye Ran

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RESEARCH INTERESTS

Medical image process, transfer learning, representation learning, and healthcare.

EDUCATION

Southwest University

M.Eng., Computer Science and Technology

Chongqing, China
Sep. 2020 — Jun. 2023

Southwest University

B.Eng., Chemical Engineering and Technology

Chongqing, China
Sep. 2016 — Jun. 2020

SELECTED RESEARCH EXPERIENCE

Feature Disentanglement for Diabetic Retinopathy

Student Research Assistant

Southwest University
Apr. 2024 — Sep. 2024

- Proposed a model to capture the evolving relationships among different grades of diabetic retinopathy.
- Designed a disentanglement loss function for VAE to decouple class-relevant features and class-irrelevant features.
- **A Method for Decoupling DR Features in UWF Fundus Images** (patent pending).

Domain Adaptation for Diabetic Retinopathy Grading

Graduate Student Researcher

Southwest University
Apr. 2023 — Mar. 2024

- Improved NT-Xent loss function and applied it to train source feature generator.
- Proposed a novel active domain adaptation method to effectively avoid outliers and fully utilize all samples.
- **Source-free Active Domain Adaptation for Diabetic Retinopathy Grading Based on Ultra-Wide-Field Fundus Image** (accepted by Comput Biol Med 2024).

Lightweight Transformer Research

Graduate Student Researcher

Southwest University
Sep. 2021 — Jun. 2023

- Designed a cosine-based matching module in Transformer encoder for aggregation of local and non-local features.
- Applied this encoder with stationary wavelet transform for super-resolution, effectively balancing model performance and parameter size.
- **Lightweight Wavelet-Based Transformer for Image Super-Resolution** (accepted by PRACAI 2022).

PUBLICATIONS

- **J. Ran**, G. Zhang, F. Xia, et al., "Source-free active domain adaptation for diabetic retinopathy grading based on ultra-wide-field fundus images," *Computers in Biology and Medicine*, vol. 2024, p. 108418, 2024. (**Q1, IF = 7.7**)
- **J. Ran** and Z. Zhang, "Lightweight Wavelet-Based Transformer for Image Super-Resolution," in *Proceedings of Pacific Rim International Conference on Artificial Intelligence*, Cham, Switzerland: Springer Nature, 2022, pp. 368-382. (**Full paper, acceptance rate = 21.1%**)
- H. Zhang, Y. Lin, **J. Ran**, et al., "Evolution stage identification of haze pollution episodes in Beijing using constrained dynamic time warping and multiway principal component analysis," *Environmental Modelling & Software*, vol. 168, p. 105811, 2023. (**Q1, IF=4.8**)

PATENTS

- W. Peng, H. Peng, J. Ge, and **J. Ran**, "Transparent instrument instance segmentation method based on CTAIS-SOLOv2," CN118097125A, Published May 28, 2024.
- W. Peng, H. Peng, M. Zhu, and **J. Ran**, "Method, device, medium and equipment for monitoring solid precipitation in transparent container," CN117058102A, Published November 14, 2023.
- H. Zhang, **J. Ran**, S. Wang, et al., "Green pricklyash peel identification method based on semantic segmentation," CN112906627B, Granted November 15, 2022.

PROJECTS

Laboratory Safety Monitoring System

Core contributor, Southwest University & Porton Pharma Solutions Ltd.

Chongqing, China

Apr. 2021 — Jan. 2022

- Developed, optimized, and deployed visual detection algorithms for unsafe behavior in laboratories.
- Built a robust live streaming backend and deployed a secure production environment using Docker.
- Achieved 12-channel video stream real-time processing on a single NVIDIA T4 GPU.

Pollutant Time Series Analysis

Experimental Lead, Southwest University

Chongqing, China

Nov. 2018 — Mar. 2019

- Conducted data preprocessing for pollution data, which improved data quality and analysis performance.
- Modeled and analyzed time series patterns to identify key pollution events.
- Applied vector error correction models to examine the relationships between pollutant concentrations and meteorological as well as economic factors.

TEACHING & INTERN

Teaching Assistant at Southwest University

Course Design of Operating System

Chongqing, China

Spring, 2021

- Collected, graded, and provided feedback on student assignments.
- Assisted students with course-related questions and provided academic support during office hours.

Machine Learning Intern

Hangzhou Xianshu Technology Co., Ltd.

Chongqing, China

Jan. 2024 — Jun. 2024

- Implemented visual recognition models for monitoring chemical reaction states.
- Developed robust instance segmentation algorithms for transparent containers in complex visual environments.

SELECTED AWARDS & HONOURS

The First Prize Scholarship for Excellence in Academic Performance

Southwest University

Dec. 2018, Dec. 2020 — Dec. 2022

The Third Prize of the 18th China Post-Graduate Mathematical Contest In Modeling

Association of Chinese Graduate Education

Dec. 2021

Outstanding Graduates

Southwest University

Apr. 2020

SKILLS

- **Languages:** Mandarin Chinese, English.
- **Programming Language:** Python, MATLAB, Linux Bash, C++.
- **Development Skills:** PyTorch, Lightning, TensorFlow, TensorRT, Latex, Docker.